

Appendix Z

Pelletized Asphalt

1. Overview. The procurement of pelletized asphalt material is to support the Airfield Damage Repair (ADR) Modernization Program. The objective of the ADR Modernization Program is to develop capabilities to rapidly repair damaged airfield pavements for the full spectrum of operational scenarios including base recovery after an attack, expedient repairs at deployed locations, and sustainment of operating surfaces at forward operating bases. In all of these scenarios, time is a critical factor, either to restore flying operations or minimize airfield closures. Of significant importance is the quality and durability of the repair or the number of aircraft passes sustained by a repaired surface before repairs must be maintained or replaced. Previous research and development activities identified new material, equipment and processes for effectively repairing craters on designated minimum aircraft operating surfaces (runways/taxiways). The technical solutions were fully validated under realistic conditions during Joint Capabilities Technology Demonstrations (JCTD) while focusing on repairing hundreds of small craters.

2. Specifications.

- a. Product contains a mixture of aggregate and pelletized asphalt mixed together.
- b. Asphalt pellets shall be well dispersed in the total aggregate gradation.
- c. Aggregate gradation shall conform to UFGS-32-12-15, paragraph 2.1 specification for dense graded asphalt.
- d. Asphalt binder and job mix formula shall conform to UFGS-32-12-15 paragraphs 2.2 and 2.3.
- e. Pelletized asphalt shall have a pellet form with particle diameter greater than .15-in and less than .35-in.
- f. Pelletized asphalt shall be coated as to be non-tacky.
- g. Pelletized asphalt shall consist of at least 88% by weight asphalt binder conforming to performance grade PG 76-22 specification as certified by the asphalt binder supplier.
- h. Pelletized asphalt shall be a quality airfield hot mix when properly mixed and heated to approximately 320°-F in the Bagela 10000 mixer or equivalent.
- i. Pelletized asphalt aggregate shall be adequately coated in approximately 10 to 20 minutes when properly heated in the Bagela 10000 mixer or equivalent.
- j. Pelletized asphalt pavement when properly compacted shall produce an asphalt pavement that meets the APA rutting depth of less than 4.5-mm as tested at 240-lb wheel load at 120-psi hose pressure using APA Standard Test Method.

3. USAF Mix Design (Aug 2016):

Product	% total mix	Lbs/50 lb total	Lbs/2000 lbs total (1 ton)
76-22 Asphalt	6.4	3.2	128
#67 (3/4")	12.2	6.1	244
#78M (3/8 ")	36.6	18.3	732
Washed Screenings	35.6	17.8	712
Dry Screenings	9.2	4.6	184
Total	100%	50 lbs	2000 lbs

(1) The above are DRY Weights. For aggregate with 5% moisture, add 5% to individual weights, for 2% moisture add 2% extra weight, etc.